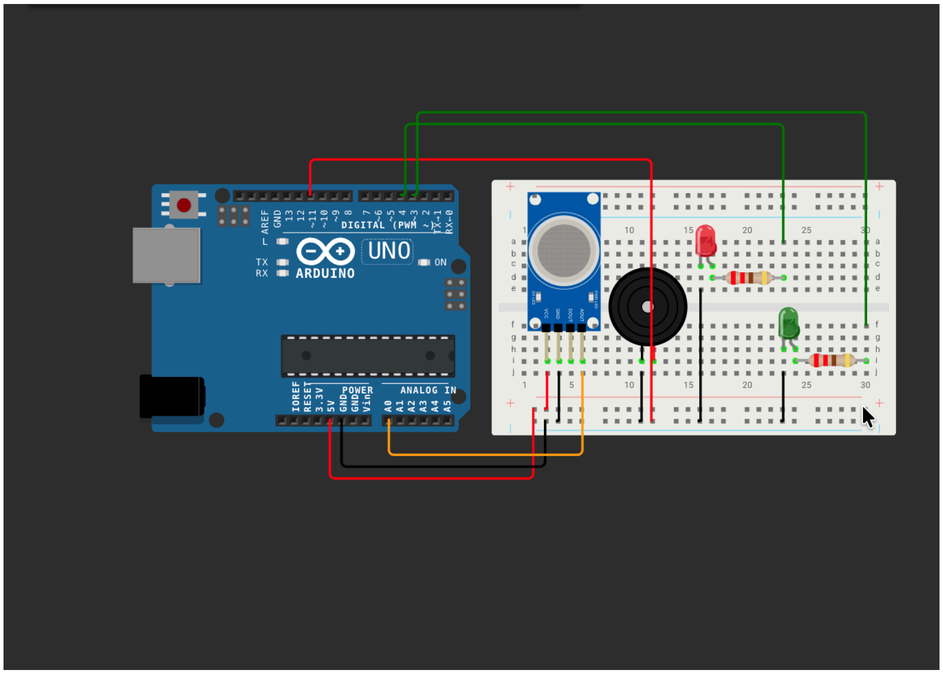
FIRE ALARM SYSTEM



Components used:

1. Arduino UNO
2. Smoke Sensor / MQ-2 Gas module
3. Red and Green LED
4. Breadboard and Wires
5. Buzzer
6. Resistor

Normal Condition

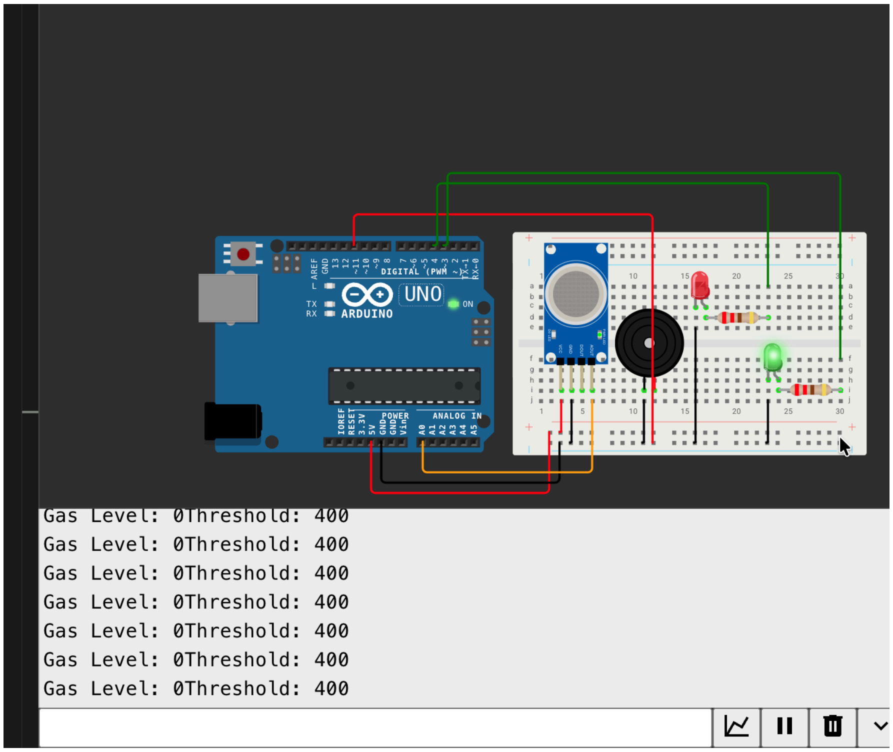
1. MQ-2 sensor detects clean air.
2. Analog value remains below threshold.
3. Arduino keeps: Green LED ON, Red LED OFF, Buzzer OFF

Fire/Smoke Detected

1. Smoke concentration increases.
2. MQ-2 sensor output voltage rises.
3. Arduino reads higher analog values
4. When value exceeds predefined threshold:Red LED turns ON, Green LED turns OFF, Buzzer sounds continuously

This alerts nearby people about potential fire hazards

Simulation



Here, the Threshold gas value is kept 400, Once the Gas level reaches above 400, Red light turns on and buzzer makes sounds to alert everyone.

Step by Step Process:

MQ-2 Gas Sensor detects gases

Arduino reads analog signals and checks the threshold

Buzzer, Light and Red LEDs are used to give audio and visual information

Breadboard, Jumper Wires, Arduino are used for power distribution